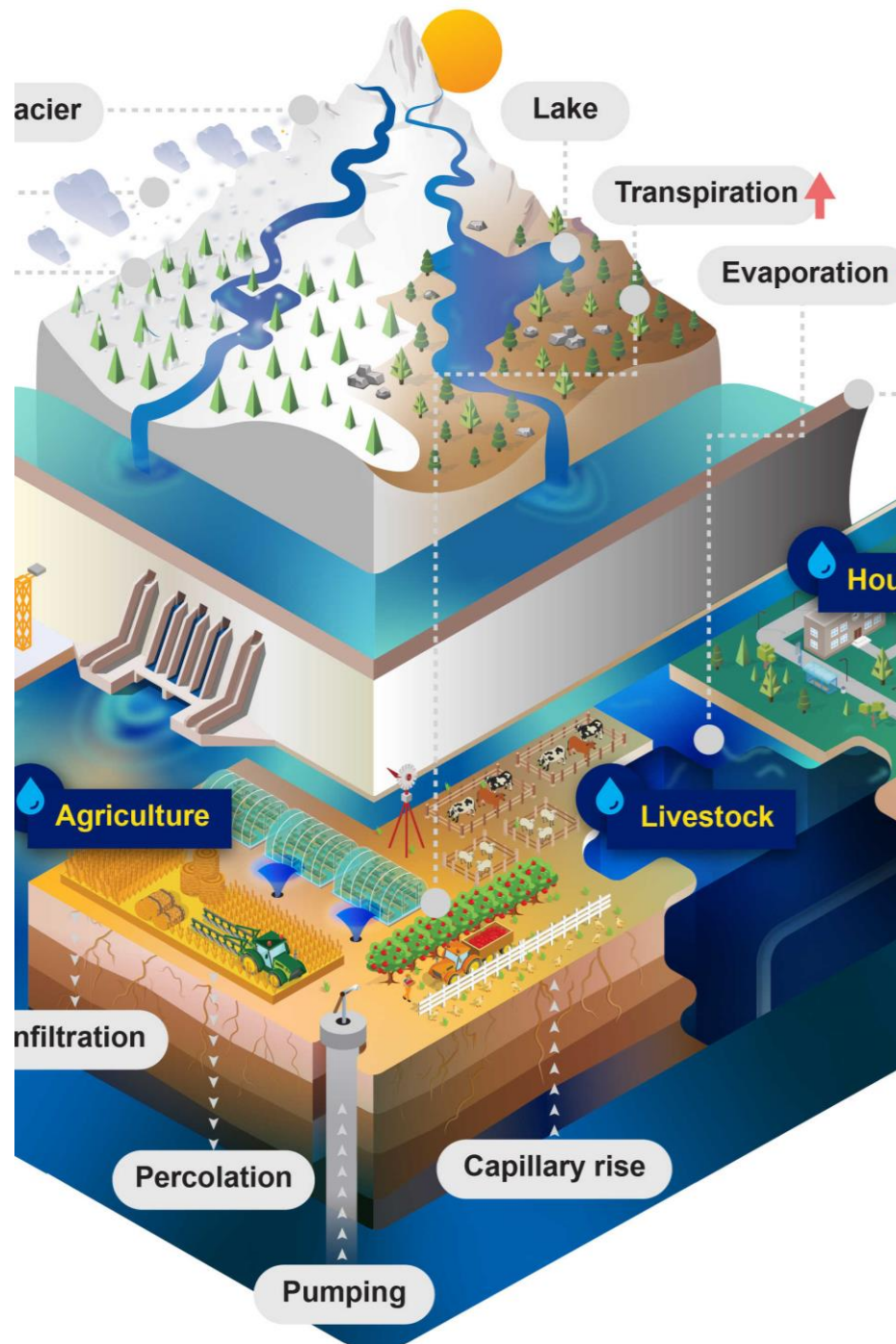


CWatM

Why GitHub?

WAT-Tech#1: CWatM in GitHub

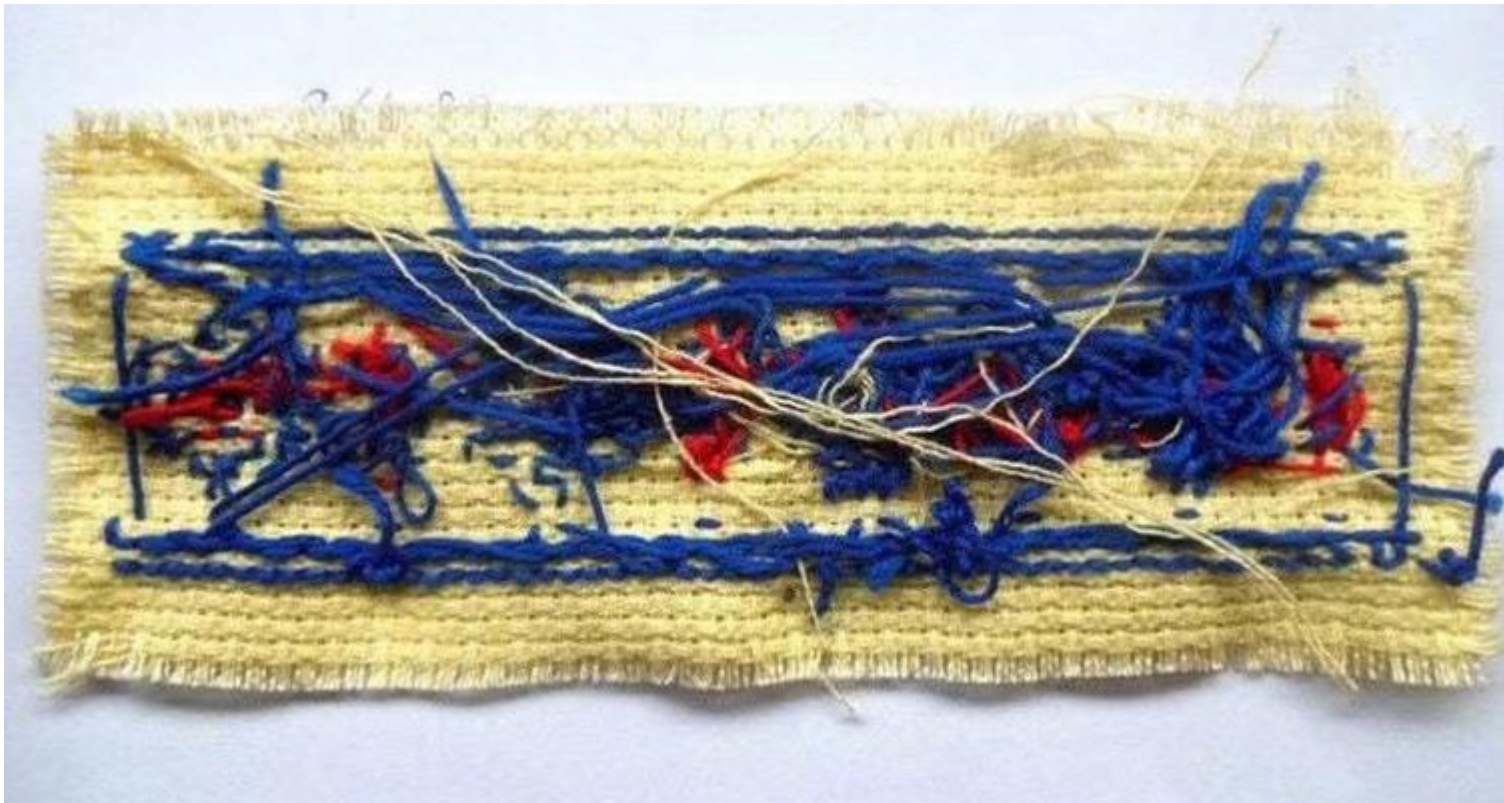
9/6/2020



Assuming this is the **frontend** of your program.



This should not be your
backend.



Stages of program development

Stage 1: first attempt for a python program displaying discharge

This PC > Local Disk (C:) > AProjects > github > model

^	Name	Date modified
PC	library_post_cwatm.py	28/05/20 09:30
PC	make_plots_discharge.py	03/06/20 14:22

Stage 2: evolving

PC > Local Disk (C:) > AProjects > github > model >

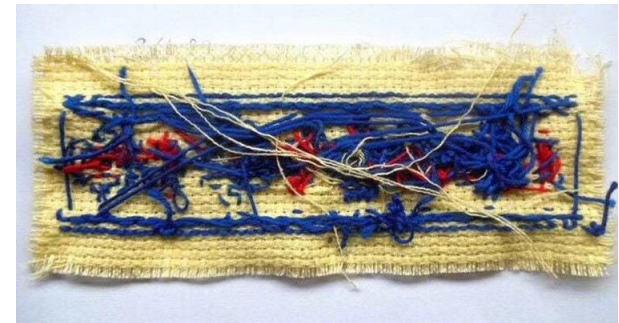
^	Name	Date modified
PC	library_post_cwatm.py	28/05/20 09:30
PC	library_post_cwatm_V2.py	28/05/20 09:30
PC	make_plots_discharge.py	03/06/20 14:22
PC	make_plots_discharge_Version2.py	03/06/20 14:22

Stages of program development

Stage 3: entropy is growing ...

PC > Local Disk (C:) > AProjects > github > model > model3

Name	Date modified
 library_post_cwatm.py	28/05/20 09:30
 library_post_cwatm_V2.py	28/05/20 09:30
 library_post_cwatm_V3.py	28/05/20 09:30
 library_post_cwatm_V3new.py	28/05/20 09:30
 make_plots_discharge.py	03/06/20 14:22
 make_plots_discharge_Version2.py	03/06/20 14:22
 make_plots_discharge_Version3.py	03/06/20 14:22
 make_plots_discharge_Version3_MS.py	03/06/20 14:22
 make_plots_discharge_Version3_MSnew.py	03/06/20 14:22
 make_plots_discharge_Version3_MSnew2.py	03/06/20 14:55
 make_plots_discharge_Version3_MSnew3.py	03/06/20 14:55
 make_plots_discharge_Version3_MSnew3new.py	03/06/20 14:55
 make_plots_discharge_Version3_MSnew3verynew.py	03/06/20 14:55



CWatM development

After around 300 development steps: ... still looks ok

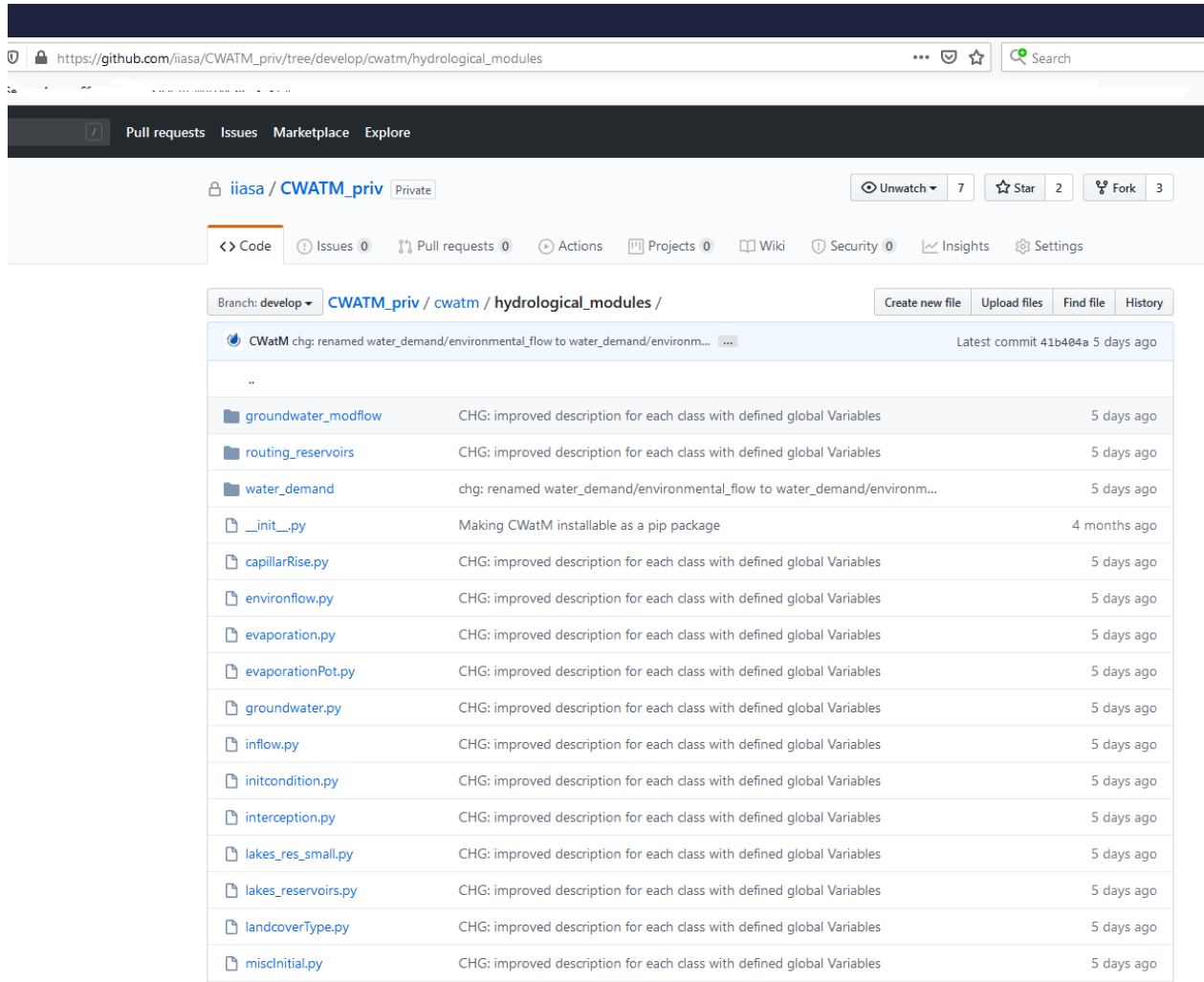
is PC > Local Disk (C:) > AProjects > github > model > cwatm

Name	Date modified
groundwater_modflow	08/06/20 12:17
routing_reservoirs	08/06/20 12:17
water_demand	08/06/20 12:17
__init__.py	16/04/20 11:54
capillarRise.py	03/06/20 12:47
environflow.py	03/06/20 12:47
evaporation.py	03/06/20 12:47
evaporationPot.py	03/06/20 12:47
groundwater.py	03/06/20 12:47
inflow.py	03/06/20 12:47
initcondition.py	03/06/20 12:47
interception.py	03/06/20 12:47
lakes_res_small.py	03/06/20 12:47
lakes_reservoirs.py	03/06/20 12:47
landcoverType.py	03/06/20 12:47
misclnitial.py	03/06/20 12:47
readmeteo.py	03/06/20 12:47
runoff_concentration.py	03/06/20 12:47
sealed_water.py	03/06/20 12:47
snow_frost.py	03/06/20 12:47
soil.py	03/06/20 12:47
waterbalance.py	03/06/20 12:47
waterquality1.py	03/06/20 12:47



CWatM development

After around 300 development steps: ... still looks ok

Branch: develop CWATM_priv / cwatm / hydrological_modules /

File	Commit Message	Time
..	..	
groundwater_modflow	CHG: improved description for each class with defined global Variables	5 days ago
routing_reservoirs	CHG: improved description for each class with defined global Variables	5 days ago
water_demand	chg: renamed water_demand/environmental_flow to water_demand/enviro...	5 days ago
__init__.py	Making CWatM installable as a pip package	4 months ago
capillarRise.py	CHG: improved description for each class with defined global Variables	5 days ago
environflow.py	CHG: improved description for each class with defined global Variables	5 days ago
evaporation.py	CHG: improved description for each class with defined global Variables	5 days ago
evaporationPot.py	CHG: improved description for each class with defined global Variables	5 days ago
groundwater.py	CHG: improved description for each class with defined global Variables	5 days ago
inflow.py	CHG: improved description for each class with defined global Variables	5 days ago
initcondition.py	CHG: improved description for each class with defined global Variables	5 days ago
interception.py	CHG: improved description for each class with defined global Variables	5 days ago
lakes_res_small.py	CHG: improved description for each class with defined global Variables	5 days ago
lakes_reservoirs.py	CHG: improved description for each class with defined global Variables	5 days ago
landcoverType.py	CHG: improved description for each class with defined global Variables	5 days ago
mislnitial.py	CHG: improved description for each class with defined global Variables	5 days ago



GitHub

What is GitHub?

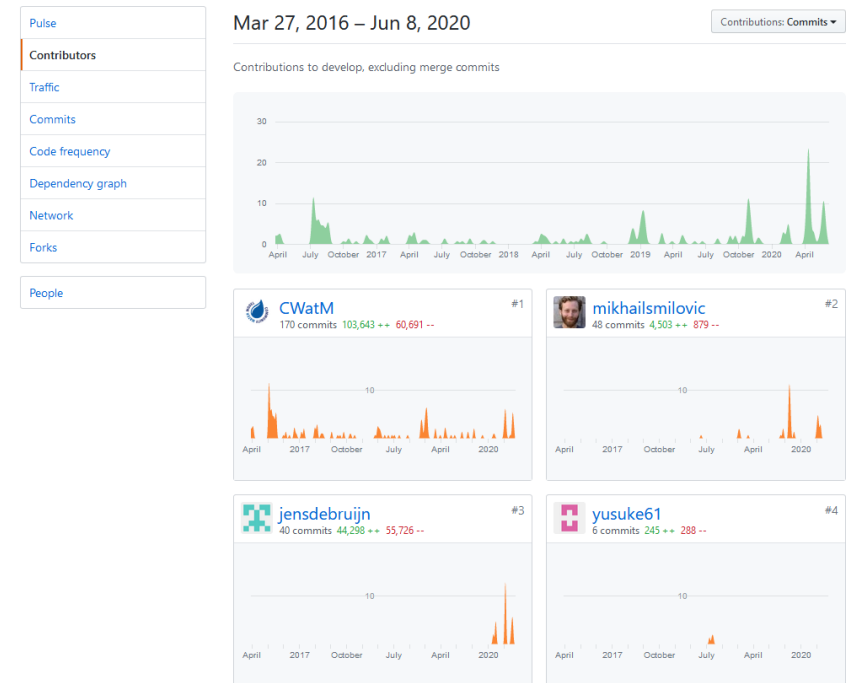
GitHub is:

- a code hosting platform
- for version control
- and collaboration.

What is GitHub?

GitHub is:

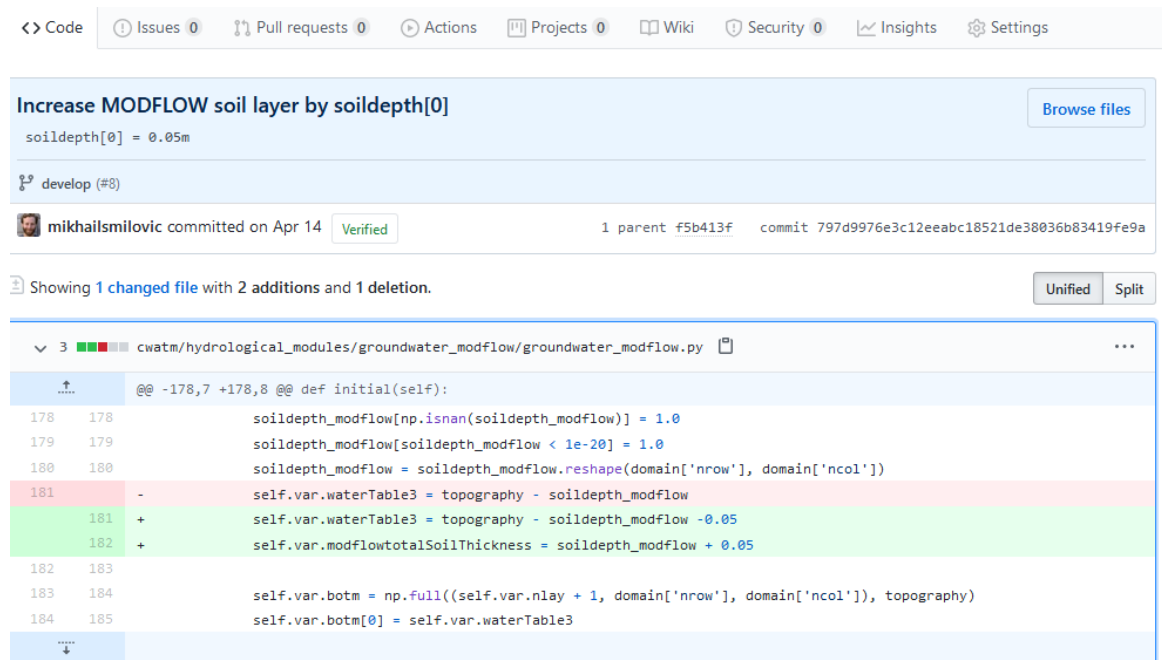
- a code hosting platform
 - A platform where you can store code or any other sort of files
 - A code sharing and publishing platform either as open source or for restricted users



What is GitHub?

GitHub is:

- a code hosting platform
- for version control
 - version control systems, manages and stores revisions of projects
 - All previous versions, changes are restorable and trackable



The screenshot shows a GitHub pull request interface. At the top, there are navigation tabs: Code, Issues (0), Pull requests (0), Actions, Projects (0), Wiki, Security (0), Insights, and Settings. The main title of the pull request is "Increase MODFLOW soil layer by soildepth[0]" with a "Browse files" button on the right. Below the title, the code snippet is shown: `soildepth[0] = 0.05m`. The branch is labeled "develop (#8)". The commit information shows "mikhailsmilovic committed on Apr 14" with a "Verified" badge, and the commit hash is "797d9976e3c12eeabc18521de38036b83419fe9a". Below this, it says "Showing 1 changed file with 2 additions and 1 deletion." and has "Unified" and "Split" view options. The file being changed is `cwatm/hydrological_modules/groundwater_modflow/groundwater_modflow.py`. The code diff shows changes to the `initial` method of a class. Line 181 has a deletion of `self.var.waterTable3 = topography - soildepth_modflow`. Line 182 has an addition of `self.var.waterTable3 = topography - soildepth_modflow - 0.05`. Line 183 has an addition of `self.var.modflowtotalSoilThickness = soildepth_modflow + 0.05`. The rest of the code remains unchanged.

What does it need to build an open-source platform

- Use of an open-source platform as a way to exchange ideas
- A high-level programming language for easy comprehension of the code and to facilitate extensibility
- A high level of modularity
- Obeying the standards of the programming language (e.g. PEP8 for Python)
- A state-of-the-art data structure for reading and writing time/spatial data
- Documentation of the model and the source code

What does it need to build an open-source platform

And nice to have:

- A semi-automatic way to facilitate documentation (e.g. Sphinx)
- A semi-automatic way to test changes of your code (e.g. pytest)
- A sort of communication with the users (e.g. forum)

Literature:

Jiménez, R. C. et al.: Four simple recommendations to encourage best practices in research software, F1000Research, 6, 10.12688/f1000research.11407.1, 2017.

Wilson, G. et al. : Best Practices for Scientific Computing, PLoS Biology, 12, 10.1371/journal.pbio.1001745, 2014.



Community Water Model (CWatM)

